

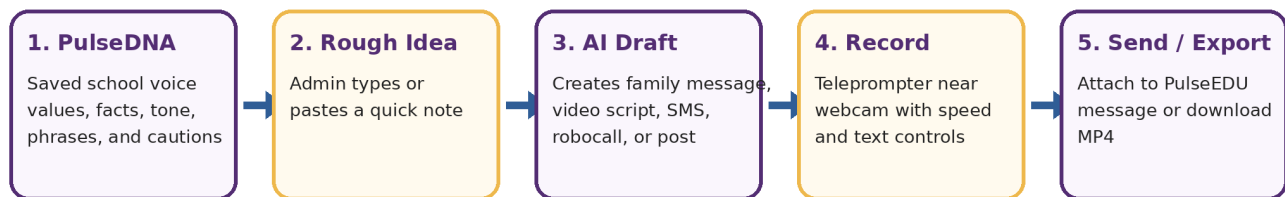
PulseEDU PulseDNA + Video Communication Studio

Build Specification for Replit Agent

Purpose: Add a script creation and video recording module to the PulseEDU family communication feature. The module turns rough ideas into polished, school-specific written posts, video scripts, and recorded family updates using a saved school communication profile called PulseDNA.

Recommended feature name: PulseDNA - Your school. Your voice. Powered by AI.

PulseDNA + Video Communication Studio Workflow



Create once, communicate everywhere - in the authentic voice of the school.

1. Product Vision

PulseEDU should become more than a message sender. This module should act as a school communication studio where an administrator can paste a rough idea, choose the audience and output type, generate a polished message, and optionally record a video with a built-in teleprompter.

2. PulseDNA: Saved School Communication Profile

PulseDNA is the persistent background context for AI-generated communication. A school can paste or upload its School Communication Profile once, save it, and use it automatically for future communications unless the user turns it off.

PulseDNA should store:

- School name, grade levels, mascot, colors, community, traditions, programs, clubs, athletics, electives, and priorities.
- Preferred tone, principal/admin voice, common phrases, greeting style, and signature language.
- Audience preferences for families, staff, students, community partners, and social media.
- Required contact information, translation expectations, accessibility notes, and do-not-say cautions.

3. Script Creation Screen

The Script Creation Screen should allow a user to paste a rough note, select an output type, select an audience and tone, then generate a polished communication using PulseDNA.

Required inputs:

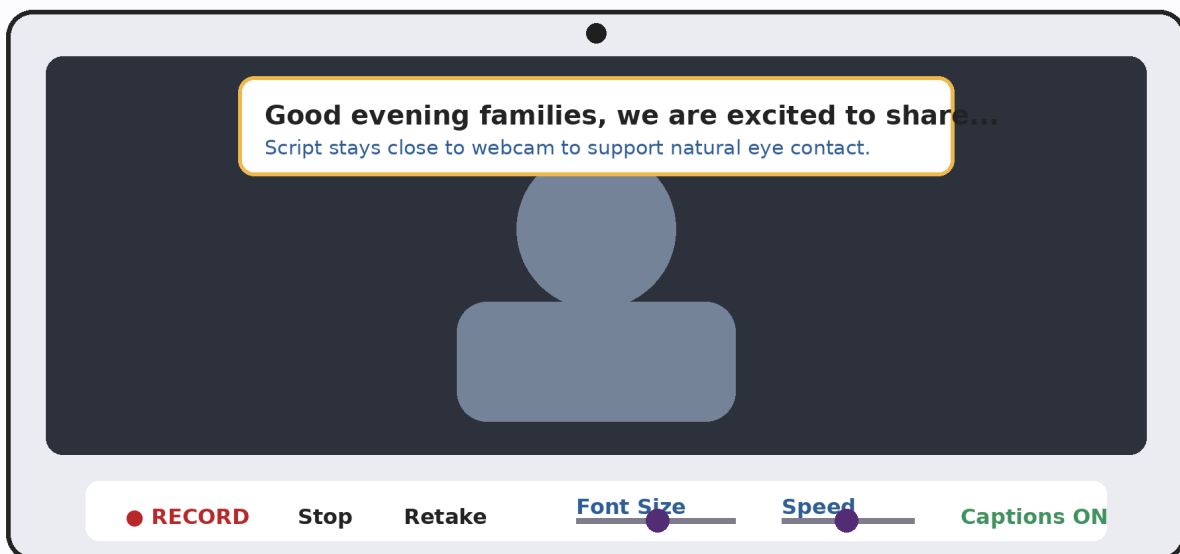
- Rough idea text box.
- Output type: Family Message, Video Script, Social Media Post, SMS/Text, Robocall Script, Staff Message, Press Release, or Newsletter Item.
- Audience: All Families, Grade Level, Specific Group, Staff, or Community.
- Tone: Warm, Urgent, Celebratory, Informational, Encouraging, or Formal.
- Language and translation options.

AI behavior:

- Use PulseDNA as the background voice and context.
- Preserve the factual intent of the rough input.
- Improve clarity, warmth, grammar, and family-friendliness.
- Do not invent dates, times, requirements, or names not provided by the user or stored in PulseDNA.
- For video scripts, write in a spoken, conversational style with natural pauses.

4. Recording Studio and Teleprompter

Recording Studio Wireframe



The camera should be centered, and the teleprompter text should sit near the top of the screen close to the laptop camera so the presenter appears to maintain eye contact.

Required controls:

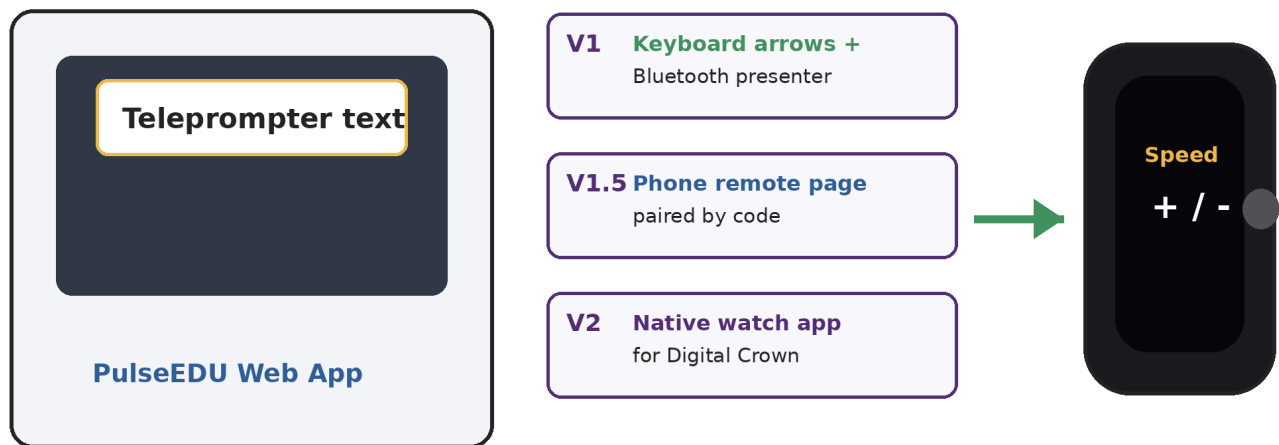
- Record, Stop, Preview, Keep Take, Delete and Record Again.
- Font size slider for the teleprompter wording.
- Scroll speed slider.
- Keyboard controls while recording: Right Arrow speeds up, Left Arrow slows down, Up Arrow increases font size, Down Arrow decreases font size.
- Spacebar pauses or resumes scrolling.
- Countdown before recording, timer while recording, and audio level indicator.

After recording:

- Preview the take.
- Erase and record another take.
- Save the accepted take.
- Attach the video to a PulseEDU family message.
- Download an MP4 for social media or another platform.
- Store accepted videos in a searchable school video library.

5. Apple Watch / Bluetooth Teleprompter Speed Control

Apple Watch / Bluetooth Control Concept



Feature request: allow the Apple Watch scroll wheel / Digital Crown or another Bluetooth control to adjust teleprompter speed while recording. This is a strong long-term idea, but it should be scoped carefully because direct Digital Crown control from a normal web app is not reliably available.

Feasibility notes:

- Direct Digital Crown access from a browser-based PulseEDU app is unlikely to be reliable because watchOS does not expose the Digital Crown as a standard laptop browser keyboard or scroll wheel input.
- A native iPhone/watch companion app is the best path if PulseEDU wants true Apple Watch control.
- Web Bluetooth exists, but it is experimental and has limited browser availability, so it should not be the primary production path for school users.
- Best V1 fallback: support common Bluetooth presentation remotes, keyboards, and media controllers that send standard key events to the browser.
- Best V1.5/V2 path: create a secure phone/watch remote that pairs with the web session by code and sends speed commands through a websocket connection.

Recommended implementation path:

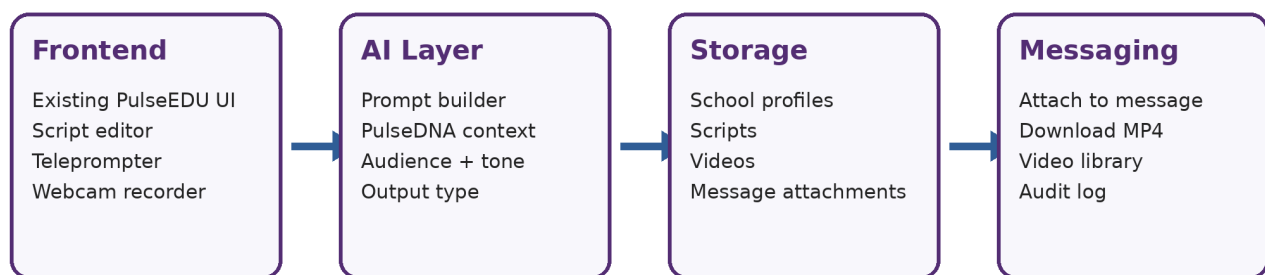
Phase	Control Method	Recommendation
V1	Keyboard + Bluetooth presenter remote	Build now. Reliable and simple.
V1.5	Phone remote web page	Pair by code and send speed controls from a phone.
V2	Native iPhone/Apple Watch companion app	Explore true Digital Crown-style control.
Avoid for V1	Direct browser-to-Apple-Watch Bluetooth	Too risky for a school production feature.

6. Accessibility and Family Communication Enhancements

- Auto-generate captions from the script or speech transcript.
- Allow captions to be embedded or exported as SRT/VTT.
- Generate translated captions and translated written versions.
- Include a readability checker for family-friendly language.
- Allow a transcript to be sent with the video for families who cannot view video.
- Include an approval workflow if the district requires review before sending.

7. Suggested Technical Architecture

Suggested Technical Architecture



Security: role-based access, school profiles, retention, captions, transcripts, and FERPA-conscious storage.

Implementation notes:

- Use browser MediaRecorder API for initial webcam capture where supported.
- Store video in secure object storage such as AWS S3, with database references tied to school, user, message, and timestamp.
- Use role-based permissions so only authorized admin/core team users can create or send messages.
- Generate AI content through a prompt builder that combines PulseDNA + rough input + selected output type + audience + tone + constraints.
- Log AI generations and final user edits for transparency and accountability.
- Provide admin settings for video retention, download permissions, and who may attach video to family messages.

8. Replit Agent Build Prompt

Add a PulseDNA-powered Script Creation and Video Communication Studio module to the existing PulseEDU family communication feature. The module should allow admin/core team users to paste or upload a school communication profile, save it as PulseDNA, and use it as persistent context for AI-generated messages. Users should be able to enter rough text, choose an output type, generate a polished message or video script, edit it, save it, and record a video using a teleprompter interface. The teleprompter should appear near the webcam, have font-size and speed sliders, support keyboard controls during recording, allow preview/retake/save, attach the completed video to a PulseEDU family message, and allow MP4 download. Include architecture for future Apple Watch/iPhone remote control, but build V1 using keyboard and Bluetooth presenter controls first.

9. Development Checklist

Area	V1 Requirements
PulseDNA	Upload/paste profile, save, edit, enable/disable, school-level storage
AI Drafting	Rough text input, output type, tone, audience, generate, revise, save
Teleprompter	Camera preview, script near webcam, font slider, speed slider, keyboard controls

Recording	Record, stop, preview, retake, save accepted take
Video Output	Attach to message, download MP4, video library entry
Accessibility	Captions/transcript, readability support, translation-ready structure
Remote Controls	Keyboard and Bluetooth presenter first; phone/watch companion as future phase

10. Source Notes for Watch/Bluetooth Feasibility

Apple documents Bluetooth device integration for native Apple Watch apps, while Web Bluetooth is described by MDN as experimental and not broadly available across major browsers. For this reason, direct browser-to-Apple-Watch Digital Crown control should be treated as future exploration, not the V1 implementation path. Practical V1 control should use keyboard events and Bluetooth presenter remotes; a native iPhone/watch companion app can be considered later for true watch control.

References reviewed: Apple Developer WWDC21 "Connect Bluetooth devices to Apple Watch"; MDN Web Bluetooth API compatibility/limited availability notes; Apple iPhone Bluetooth accessory support documentation.